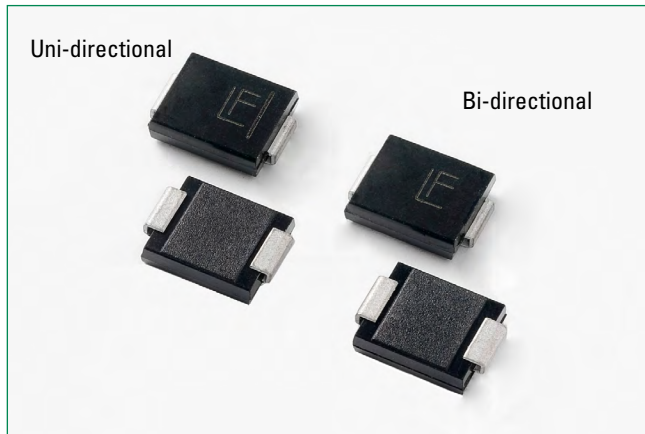


# TPSMC-VR Series

## Surface Mount – 1500W



### Agency Approvals

| Agency | Agency File Number |
|--------|--------------------|
|        | E230531            |

### Maximum Ratings and Thermal Characteristics

( $T_A = 25^\circ\text{C}$  unless otherwise noted)

| Parameter                                                                                            | Symbol          | Value      | Unit               |
|------------------------------------------------------------------------------------------------------|-----------------|------------|--------------------|
| Peak Pulse Power Dissipation (IPP x VC) by 10/1000 $\mu\text{s}$ Waveform (Fig.2) (Note 1), (Note 2) | $P_{PPM}$       | 1500       | W                  |
| Power Dissipation on Infinite Heat Sink at $T_A = 50^\circ\text{C}$                                  | $P_{M(AV)}$     | 6.5        | W                  |
| Peak Forward Surge Current, 8.3ms Single Half Sine Wave                                              | $I_{FSM}$       | 200        | A                  |
| Maximum Instantaneous Forward Voltage at 100A for Unidirectional Only (Note 4)                       | $V_F$           | 3.5        | V                  |
| Operating Junction and Storage Temperature Range                                                     | $T_J, T_{STG}$  | -65 to 150 | $^\circ\text{C}$   |
| Typical Thermal Resistance Junction to Lead                                                          | $R_{\theta JL}$ | 15         | $^\circ\text{C/W}$ |
| Typical Thermal Resistance Junction to Ambient                                                       | $R_{\theta JA}$ | 75         | $^\circ\text{C/W}$ |

#### Notes:

- Non-repetitive current pulse per Fig. 4 and derated above  $T_A = 25^\circ\text{C}$  per Fig. 3.
- Mounted on copper pad area of 0.31x0.31" (8.0 x 8.0mm) to each terminal.
- Measured on 8.3ms single half sine wave or equivalent square wave for unidirectional component only, duty cycle=4 per minute maximum.

### Description

The TPSMC-VR series is designed specifically to protect sensitive electronic equipment from voltage transients induced by lightning and other transient voltage events.

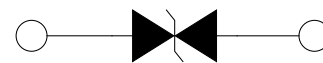
### Features & Benefits

- High reliability application and automotive grade AEC-Q101 qualified
- Surface mount component to optimize board space
- Low profile package.
- Typical failure mode is short from over-specified voltage or current
- Whisker test is conducted based on JEDEC JESD201A per its table 4a and 4c
- ESD protection of data lines in accordance with IEC 61000-4-2, 30kV(Air), 30kV (Contact)
- EFT protection of data lines in accordance with IEC 61000-4-4
- Built-in strain relief
- Glass passivated chip junction
- 1500W peak pulse power capability at 10/1000 $\mu\text{s}$  waveform, repetition rate (duty cycles): 0.01 %
- Fast response time: typically less than 1.0ps from 0V to VBR min
- Excellent clamping capability
- Low incremental surge resistance
- UL Recognized compound meeting flammability rating V-0.
- Meet MSL level1, per J-STD-020, High temperature soldering guaranteed: 260 $^\circ\text{C}$ /10 seconds at terminals
- Matte tin lead-free plated
- Halogen free and RoHS compliant
- Pb-free E3 means 2nd level interconnect is Pb-free and the terminal finish material is tin(Sn) (IPC/JEDEC J-STD-609A.01)

### Applications

TVS components are ideal for the protection of I/O interfaces,  $V_{CC}$  bus and other vulnerable circuits used in automotive applications.

### Functional Diagram



Bi-directional



Uni-directional

# TPSMC-VR Series

## Surface Mount – 1500W

### Electrical Characteristics

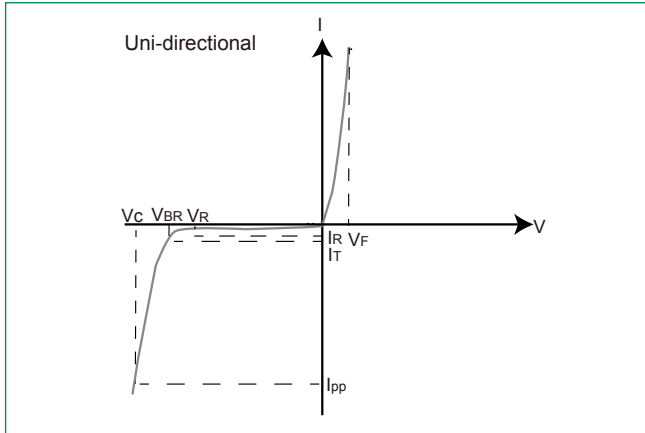
| Part Number (Uni) | Part Number (Bi) | Marking |      | Reverse Stand off Voltage $V_R$ (Volts) | Breakdown Voltage $V_{BR}$ (Volts) @ $I_T$ |        | Test Current $I_T$ (mA) | Maximum Clamping Voltage $V_C$ @ $I_{PP}$ (V) | Maximum Peak Pulse Current $I_{PP}$ (A) | Maximum Reverse Leakage $I_R$ @ $V_R$ ( $\mu$ A) | Maximum Temperature coefficient of $V_{BR}$ (%/C) | Agency Approval  |
|-------------------|------------------|---------|------|-----------------------------------------|--------------------------------------------|--------|-------------------------|-----------------------------------------------|-----------------------------------------|--------------------------------------------------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------|
|                   |                  | UNI     | BI   |                                         | MIN                                        | MAX    |                         |                                               |                                         |                                                  |                                                   |                                                                                                     |
| TPSMC11A-VR       | TPSMC11CA-VR     | GDZA    | BDZA | 11.0                                    | 12.20                                      | 13.50  | 1                       | 18.2                                          | 82.5                                    | 1                                                | 0.074                                             | X                                                                                                   |
| TPSMC12A-VR       | TPSMC12CA-VR     | GEEA    | BEEA | 12.0                                    | 13.30                                      | 14.70  | 1                       | 19.9                                          | 75.4                                    | 1                                                | 0.075                                             | X                                                                                                   |
| TPSMC13A-VR       | TPSMC13CA-VR     | GEGA    | BEGA | 13.0                                    | 14.40                                      | 15.90  | 1                       | 21.5                                          | 69.8                                    | 1                                                | 0.076                                             | X                                                                                                   |
| TPSMC14A-VR       | TPSMC14CA-VR     | GEKA    | BEKA | 14.0                                    | 15.60                                      | 17.20  | 1                       | 23.2                                          | 64.7                                    | 1                                                | 0.080                                             |                                                                                                     |
| TPSMC15A-VR       | TPSMC15CA-VR     | GEMA    | BEMA | 15.0                                    | 16.70                                      | 18.50  | 1                       | 24.4                                          | 61.5                                    | 1                                                | 0.083                                             | X                                                                                                   |
| TPSMC16A-VR       | TPSMC16CA-VR     | GEPA    | BEPa | 16.0                                    | 17.80                                      | 19.70  | 1                       | 26.0                                          | 57.7                                    | 1                                                | 0.084                                             | X                                                                                                   |
| TPSMC17A-VR       | TPSMC17CA-VR     | GERA    | BERA | 17.0                                    | 18.90                                      | 20.90  | 1                       | 27.6                                          | 54.4                                    | 1                                                | 0.085                                             |                                                                                                     |
| TPSMC18A-VR       | TPSMC18CA-VR     | GETA    | BETA | 18.0                                    | 20.00                                      | 22.10  | 1                       | 29.2                                          | 51.4                                    | 1                                                | 0.088                                             | X                                                                                                   |
| TPSMC20A-VR       | TPSMC20CA-VR     | GEVA    | BEVA | 20.0                                    | 22.20                                      | 24.50  | 1                       | 32.4                                          | 46.3                                    | 1                                                | 0.091                                             | X                                                                                                   |
| TPSMC22A-VR       | TPSMC22CA-VR     | GEXA    | BEXA | 22.0                                    | 24.40                                      | 26.90  | 1                       | 35.5                                          | 42.3                                    | 1                                                | 0.092                                             | X                                                                                                   |
| TPSMC24A-VR       | TPSMC24CA-VR     | GEZA    | BEZA | 24.0                                    | 26.70                                      | 29.50  | 1                       | 38.9                                          | 38.6                                    | 1                                                | 0.092                                             | X                                                                                                   |
| TPSMC26A-VR       | TPSMC26CA-VR     | GFEA    | BFEA | 26.0                                    | 28.90                                      | 31.90  | 1                       | 42.1                                          | 35.7                                    | 1                                                | 0.093                                             |                                                                                                     |
| TPSMC28A-VR       | TPSMC28CA-VR     | GFGA    | BFGA | 28.0                                    | 31.10                                      | 34.40  | 1                       | 45.4                                          | 33.1                                    | 1                                                | 0.094                                             |                                                                                                     |
| TPSMC30A-VR       | TPSMC30CA-VR     | GFKa    | BFKA | 30.0                                    | 33.30                                      | 36.80  | 1                       | 48.4                                          | 31.0                                    | 1                                                | 0.096                                             | X                                                                                                   |
| TPSMC33A-VR       | TPSMC33CA-VR     | GFMA    | BFMA | 33.0                                    | 36.70                                      | 40.60  | 1                       | 53.3                                          | 28.2                                    | 1                                                | 0.097                                             | X                                                                                                   |
| TPSMC36A-VR       | TPSMC36CA-VR     | GFPA    | BFPA | 36.0                                    | 40.00                                      | 44.20  | 1                       | 58.1                                          | 25.9                                    | 1                                                | 0.098                                             | X                                                                                                   |
| TPSMC40A-VR       | TPSMC40CA-VR     | GFRA    | BFRA | 40.0                                    | 44.40                                      | 49.10  | 1                       | 64.5                                          | 23.3                                    | 1                                                | 0.099                                             |                                                                                                     |
| TPSMC43A-VR       | TPSMC43CA-VR     | GFTA    | BFTA | 43.0                                    | 47.80                                      | 52.80  | 1                       | 69.4                                          | 21.7                                    | 1                                                | 0.100                                             | X                                                                                                   |
| TPSMC45A-VR       | TPSMC45CA-VR     | GFVA    | BFVA | 45.0                                    | 50.00                                      | 55.30  | 1                       | 72.7                                          | 20.6                                    | 1                                                | 0.101                                             |                                                                                                     |
| TPSMC48A-VR       | TPSMC48CA-VR     | GFXA    | BFXA | 48.0                                    | 53.30                                      | 58.90  | 1                       | 77.4                                          | 19.4                                    | 1                                                | 0.101                                             |                                                                                                     |
| TPSMC51A-VR       | TPSMC51CA-VR     | GFZA    | BFZA | 51.0                                    | 56.70                                      | 62.70  | 1                       | 82.4                                          | 18.2                                    | 1                                                | 0.101                                             | X                                                                                                   |
| TPSMC54A-VR       | TPSMC54CA-VR     | GGEA    | BGEA | 54.0                                    | 60.00                                      | 66.30  | 1                       | 87.1                                          | 17.3                                    | 1                                                | 0.102                                             |                                                                                                     |
| TPSMC58A-VR       | TPSMC58CA-VR     | GGGA    | BGGA | 58.0                                    | 64.40                                      | 71.20  | 1                       | 93.6                                          | 16.1                                    | 1                                                | 0.103                                             |                                                                                                     |
| TPSMC60A-VR       | TPSMC60CA-VR     | GGKA    | BGKA | 60.0                                    | 66.70                                      | 73.70  | 1                       | 96.8                                          | 15.5                                    | 1                                                | 0.103                                             |                                                                                                     |
| TPSMC64A-VR       | TPSMC64CA-VR     | GGMA    | BGMA | 64.0                                    | 71.10                                      | 78.60  | 1                       | 103.0                                         | 14.6                                    | 1                                                | 0.104                                             |                                                                                                     |
| TPSMC70A-VR       | TPSMC70CA-VR     | GGPA    | BGPA | 70.0                                    | 77.80                                      | 86.00  | 1                       | 113.0                                         | 13.3                                    | 1                                                | 0.105                                             |                                                                                                     |
| TPSMC75A-VR       | TPSMC75CA-VR     | GGRA    | BGRA | 75.0                                    | 83.30                                      | 92.10  | 1                       | 121.0                                         | 12.4                                    | 1                                                | 0.106                                             | X                                                                                                   |
| TPSMC78A-VR       | TPSMC78CA-VR     | GGTA    | BGTA | 78.0                                    | 86.70                                      | 95.80  | 1                       | 126.0                                         | 11.9                                    | 1                                                | 0.106                                             |                                                                                                     |
| TPSMC85A-VR       | TPSMC85CA-VR     | GGVA    | BGVA | 85.0                                    | 94.40                                      | 104.00 | 1                       | 137.0                                         | 11.0                                    | 1                                                | 0.106                                             |                                                                                                     |
| TPSMC90A-VR       | TPSMC90CA-VR     | GHEA    | BHEA | 90                                      | 100                                        | 111    | 1                       | 145.4                                         | 10.3                                    | 1                                                | 0.107                                             |                                                                                                     |
| TPSMC100A-VR      | TPSMC100CA-VR    | GHFA    | BHFA | 100                                     | 111                                        | 123    | 1                       | 159.9                                         | 9.4                                     | 1                                                | 0.107                                             | X                                                                                                   |
| TPSMC110A-VR      | TPSMC110CA-VR    | GHGA    | BHGA | 110                                     | 122                                        | 135    | 1                       | 175.5                                         | 8.6                                     | 1                                                | 0.107                                             | X                                                                                                   |
| TPSMC120A-VR      | TPSMC120CA-VR    | GHHA    | BHHA | 120                                     | 133                                        | 147    | 1                       | 191.1                                         | 7.9                                     | 1                                                | 0.108                                             | X                                                                                                   |
| TPSMC130A-VR      | TPSMC130CA-VR    | GHIA    | BHIA | 130                                     | 144                                        | 159    | 1                       | 206.7                                         | 7.3                                     | 1                                                | 0.108                                             | X                                                                                                   |
| TPSMC150A-VR      | TPSMC150CA-VR    | GHKA    | BHKA | 150                                     | 167                                        | 185    | 1                       | 240.5                                         | 6.2                                     | 1                                                | 0.108                                             | X                                                                                                   |
| TPSMC160A-VR      | TPSMC160CA-VR    | GHLA    | BHLA | 160                                     | 178                                        | 197    | 1                       | 256.1                                         | 5.9                                     | 1                                                | 0.108                                             | X                                                                                                   |
| TPSMC170A-VR      | TPSMC170CA-VR    | GHMA    | BHMA | 170                                     | 189                                        | 209    | 1                       | 271.7                                         | 5.5                                     | 1                                                | 0.108                                             | X                                                                                                   |
| TPSMC180A-VR      | TPSMC180CA-VR    | GHNA    | BHNA | 180                                     | 201                                        | 222    | 1                       | 288.6                                         | 5.2                                     | 1                                                | 0.108                                             | X                                                                                                   |
| TPSMC188A-VR      | TPSMC188CA-VR    | GHOA    | BHOA | 188                                     | 209                                        | 231    | 1                       | 300.3                                         | 5.0                                     | 1                                                | 0.11                                              |                                                                                                     |
| TPSMC200A-VR      | TPSMC200CA-VR    | GHPA    | BHPA | 200                                     | 224                                        | 247    | 1                       | 321.1                                         | 4.7                                     | 1                                                | 0.11                                              | X                                                                                                   |
| TPSMC220A-VR      | TPSMC220CA-VR    | GHXA    | BHXA | 220                                     | 246                                        | 272    | 1                       | 356                                           | 4.2                                     | 1                                                | 0.12                                              | X                                                                                                   |
| TPSMC250A-VR      | TPSMC250CA-VR    | GHZA    | BHZA | 250                                     | 279                                        | 309    | 1                       | 405                                           | 3.7                                     | 1                                                | 0.12                                              | X                                                                                                   |
| TPSMC300A-VR      | TPSMC300CA-VR    | GJEA    | BJEA | 300                                     | 335                                        | 371    | 1                       | 486                                           | 3.1                                     | 1                                                | 0.12                                              | X                                                                                                   |
| TPSMC350A-VR      | TPSMC350CA-VR    | GJGA    | BJGA | 350                                     | 391                                        | 432    | 1                       | 567                                           | 2.6                                     | 1                                                | 0.12                                              | X                                                                                                   |
| TPSMC400A-VR      | TPSMC400CA-VR    | GJKA    | BJKA | 400                                     | 447                                        | 494    | 1                       | 648                                           | 2.3                                     | 1                                                | 0.12                                              | X                                                                                                   |
| TPSMC440A-VR      | TPSMC440CA-VR    | GJMA    | BJMA | 440                                     | 492                                        | 543    | 1                       | 713                                           | 2.1                                     | 1                                                | 0.12                                              | X                                                                                                   |

$V_{BR} @ T_J = V_{BR} @ 25^\circ\text{C} \times (1 + \alpha T \times (T_J - 25))$  ( $\alpha$ : Temperature Coefficient, typical value is 0.1 %)

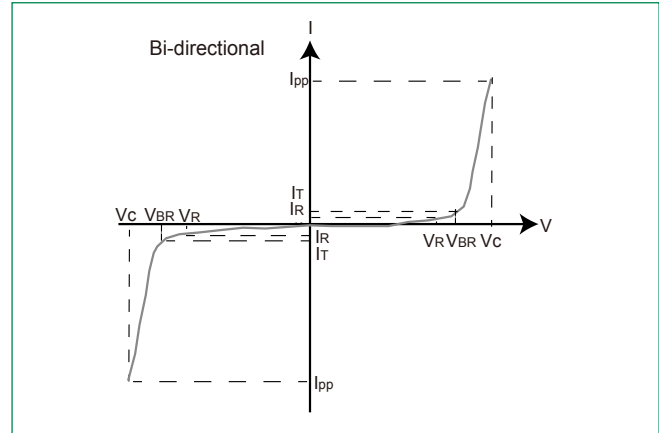
# TPSMC-VR Series

## Surface Mount – 1500W

### I-V Curve Characteristics



**$P_{PPM}$  Peak Pulse Power Dissipation** ( $I_{PP} \times V_C$ ) – Max power dissipation  
 **$V_R$  Stand-off Voltage** – Maximum voltage that can be applied to the TVS without operation  
 **$V_{BR}$  Breakdown Voltage** – Maximum voltage that flows through the TVS at a specified test current ( $I_T$ )



**$V_C$  Clamping Voltage** – Peak voltage measured across the TVS at a specified  $I_{PP}$  (peak impulse current)  
 **$I_R$  Reverse Leakage Current** – Current measured at  $V_R$   
 **$V_F$  Forward Voltage Drop for Uni-directional**

### Ratings and Characteristic Curves ( $T_A=25^\circ\text{C}$ unless otherwise noted)

Figure 1: TVS Transients Clamping Waveform

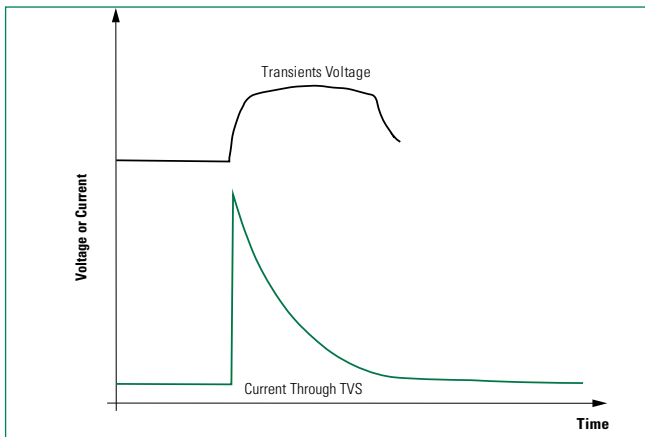


Figure 2: Peak Pulse Power Rating

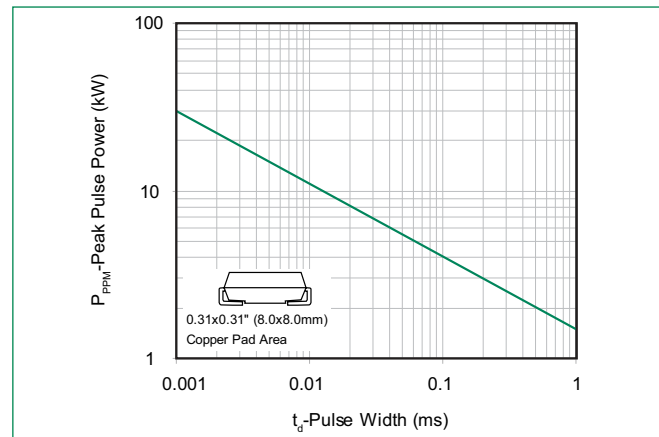


Figure 3: Peak Pulse Power Derating Curve

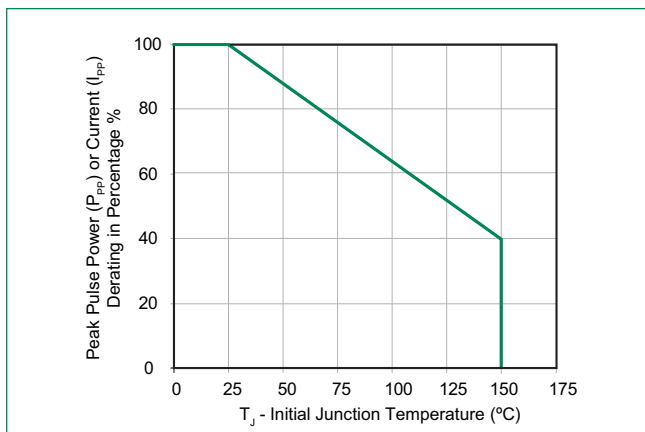
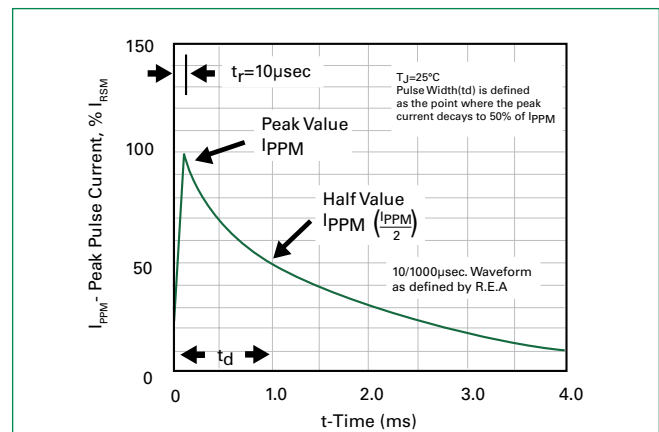


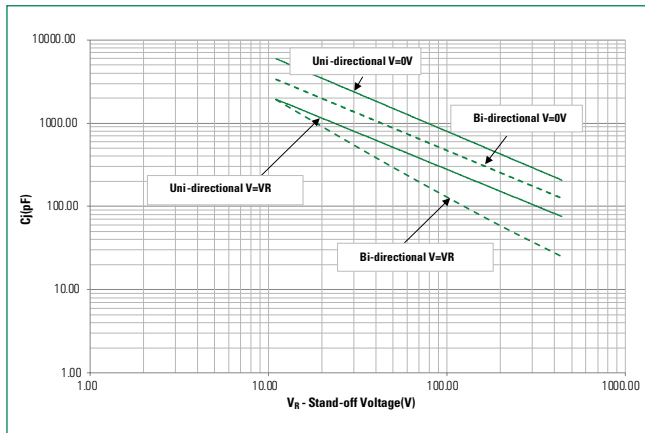
Figure 4: Pulse Waveform



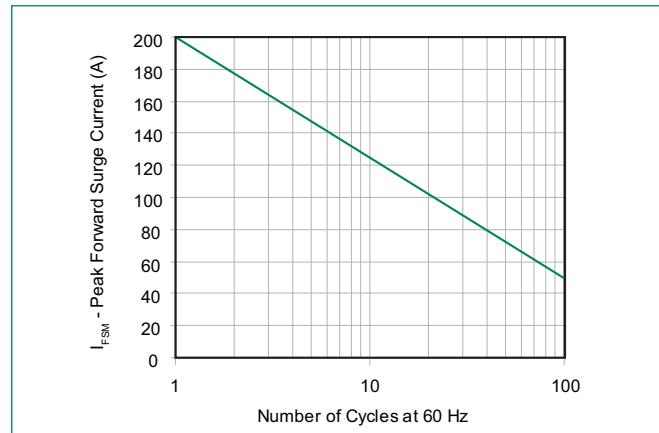
# TPSMC-VR Series

## Surface Mount – 1500W

**Figure 5 - Typical Junction Capacitance**

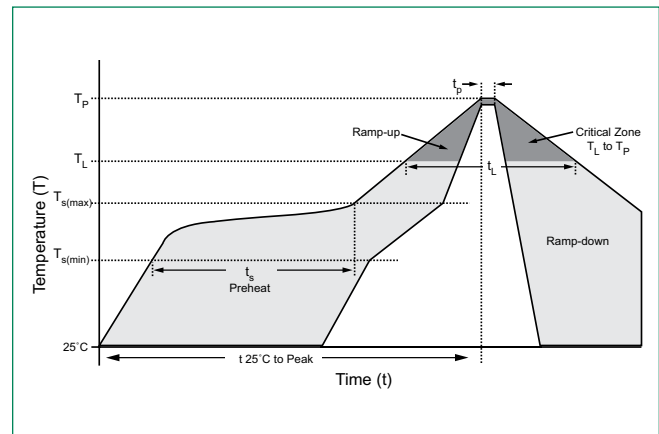


**Figure 6 - Maximum Non-Repetitive Peak Forward Surge Current Uni-Directional Only**



## Soldering Parameters

|                                                                        |                                    |                    |
|------------------------------------------------------------------------|------------------------------------|--------------------|
| <b>Reflow Condition</b>                                                |                                    | Lead-free assembly |
| <b>Pre Heat</b>                                                        | - Temperature Min ( $T_{s(min)}$ ) | 150°C              |
|                                                                        | - Temperature Max ( $T_{s(max)}$ ) | 200°C              |
|                                                                        | - Time (min to max) ( $t_s$ )      | 60 – 120 secs      |
| <b>Average ramp up rate (Liquidus Temp (<math>T_L</math>) to peak)</b> |                                    | 3°C/second max     |
| <b><math>T_{s(max)}</math> to <math>T_L</math> - Ramp-up Rate</b>      |                                    | 3°C/second max     |
| <b>Reflow</b>                                                          | - Temperature ( $T_L$ ) (Liquidus) | 217°C              |
|                                                                        | - Time (min to max) ( $t_s$ )      | 60 – 150 seconds   |
| <b>Peak Temperature (<math>T_p</math>)</b>                             |                                    | 260 $\pm$ 0.5 °C   |
| <b>Time within 5°C of actual peak Temperature (<math>t_p</math>)</b>   |                                    | 30 seconds max     |
| <b>Ramp-down Rate</b>                                                  |                                    | 6°C/second max     |
| <b>Time 25°C to peak Temperature (<math>T_p</math>)</b>                |                                    | 8 minutes max.     |
| <b>Do not exceed</b>                                                   |                                    | 260°C              |



## Physical Specifications

|                 |                                                                   |
|-----------------|-------------------------------------------------------------------|
| <b>Weight</b>   | 0.007 ounce, 0.21 grams                                           |
| <b>Case</b>     | JEDEC DO214AB. Molded plastic body over glass passivated junction |
| <b>Polarity</b> | Color band denotes cathode for unidirectional components          |
| <b>Terminal</b> | Matte Tin-plated leads, Solderable per JESD22-B102                |

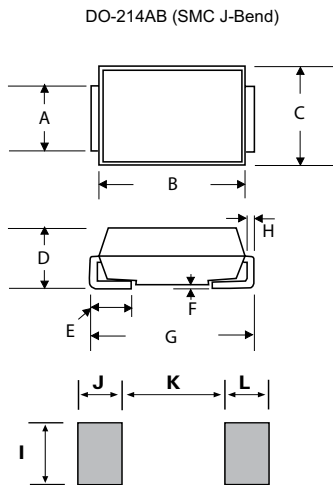
## Environmental Specifications

|                            |                          |
|----------------------------|--------------------------|
| <b>High Temp. Storage</b>  | JESD22-A103              |
| <b>HTRB</b>                | JESD22-A108              |
| <b>Temperature Cycling</b> | JESD22-A104              |
| <b>MSL</b>                 | JEDEC-J-STD-020, Level 1 |
| <b>H3TRB</b>               | JESD22-A101              |
| <b>RSH</b>                 | JESD22-A111              |

# TPSMC-VR Series

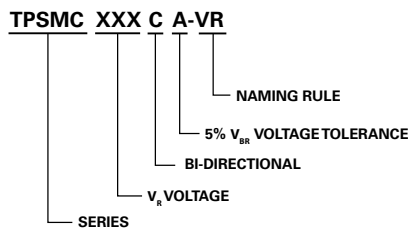
## Surface Mount – 1500W

### Dimensions

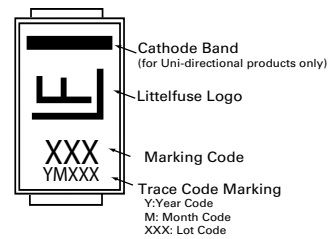


| Dimensions | Inches |       | Millimeters |       |
|------------|--------|-------|-------------|-------|
|            | Min    | Max   | Min         | Max   |
| <b>A</b>   | 0.114  | 0.126 | 2.900       | 3.200 |
| <b>B</b>   | 0.260  | 0.280 | 6.600       | 7.110 |
| <b>C</b>   | 0.220  | 0.245 | 5.590       | 6.220 |
| <b>D</b>   | 0.079  | 0.103 | 2.060       | 2.620 |
| <b>E</b>   | 0.030  | 0.060 | 0.760       | 1.520 |
| <b>F</b>   | -      | 0.008 | -           | 0.203 |
| <b>G</b>   | 0.305  | 0.320 | 7.750       | 8.130 |
| <b>H</b>   | 0.006  | 0.012 | 0.152       | 0.305 |
| <b>I</b>   | 0.129  | -     | 3.300       | -     |
| <b>J</b>   | 0.094  | -     | 2.400       | -     |
| <b>K</b>   | -      | 0.165 | -           | 4.200 |
| <b>L</b>   | 0.094  | -     | 2.400       | -     |

### Part Numbering System



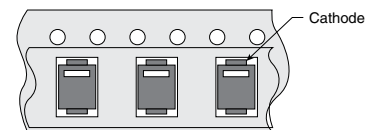
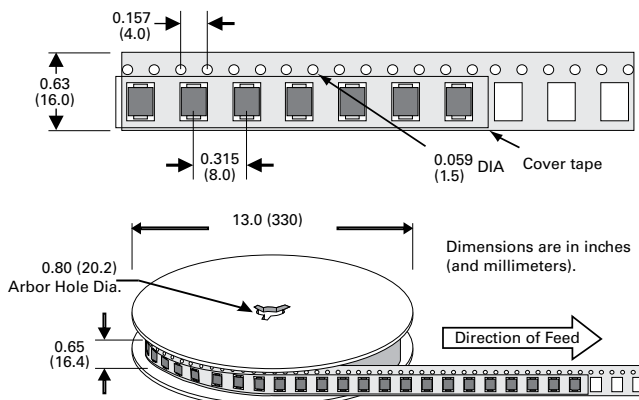
### Part Marking System



### Packaging

| Part number   | Component Package | Quantity | Packaging Option                 | Packaging Specification |
|---------------|-------------------|----------|----------------------------------|-------------------------|
| TPSMCxxxXX-VR | DO-214AB          | 3000     | Tape & Reel - 16mm tape/13" reel | EIA STD RS-481          |

### Tape and Reel Specification



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